Sushan Sapkota

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OBJECTIVE

Motivated and detail-oriented Mechanical Engineering student at Howard University seeking an engineering internship to apply academic knowledge in dynamics, design analysis, MATLAB, manufacturing, robotics, and optimization to real-world engineering challenges. Eager to contribute strong problem-solving skills, hands-on technical aptitude, and collaborative teamwork.

EDUCATION

Howard University - Washington, D.C.

Bachelor of Science in Mechanical Engineering, Expected May 2027

Relevant Coursework: Statics, Dynamics, Thermodynamics, Mechanics of Materials, Computer-Aided Design (CAD), MATLAB, Optimization, Manufacturing, Robotics

College of Southern Maryland - La Plata, MD | Associate of Science in Engineering (Attended until May 2025)

Dean's List (Fall 2024, Spring 2024), GPA: 4.0

Idaho State University - Pocatello, ID | Coursework toward B.S. in Mechanical Engineering (Attended until December 2023)

TECHNICAL SKILLS

- **Software & Design:** MATLAB, AutoCAD, SolidWorks, CAD Modeling, MS Office Suite
- **Manufacturing & Robotics:** Manufacturing Processes, Robotics Fundamentals, Process Optimization, Design for Manufacturing
- **Engineering & Analysis:** Engineering Optimization, Design Analysis, Test Planning, Stress-Strain Analysis, Material Property Evaluation
- **Technical Abilities:** Troubleshooting, Minor Maintenance & Repair, Hands-on Prototyping, Quality Assurance, Technical Documentation

PROJECT EXPERIENCE

Solid Mechanics - Cantilever Beam Deflection Project | Howard University, Spring 2026

- Conducted experimental cantilever-beam testing using 10-100 g incremental loads and dial-gauge measurements to analyze load-deflection behavior.
- Applied beam-deflection and bending-stress theory to determine Young's modulus of **442,694 psi (3.05 GPa)** and maximum bending stress of **560.56 psi**.
- Performed linear regression of experimental data (**$R^2 \approx 0.994$**) and compared material properties with published values, identifying Balsa wood as the closest match with **10.29% error**.

Engineering Design Project | Howard University, Fall 2025

- Collaborated with a 4-member team to design an energy-efficient structural component using SolidWorks.
- Applied MATLAB for engineering calculations and validation of design specifications, improving design reliability.
- Presented the final design and findings to a faculty panel, demonstrating technical communication and teamwork.

PROFESSIONAL EXPERIENCE

Clerk - Jay's Market, Waldorf, MD | Jan 2024 - Dec 2024

- Managed inventory and maintained organized workflows to support efficient daily operations.
- Delivered customer service and resolved inquiries while balancing multiple tasks in a fast-paced environment.

Custodial Assistant - Idaho State University, Pocatello, ID | Aug 2023 - Jan 2024

- Performed minor equipment maintenance and troubleshooting to reduce operational downtime.
- Maintained campus safety and cleanliness and responded quickly to emergent operational issues.

ADDITIONAL INFORMATION

Languages: English (Professional Proficiency), Nepali (Native/Bilingual) | **Interests:** Technology, Hands-on Prototyping, Camping, Sports